

CF-MARLOS: Cooperative Federated Multi-Agent Reinforcement Learning with Learnable Aggregation for Open-Set Blockchain Intrusion Detection

Mohammad Hassan Amini

M.Sc. Student, Iran University of Science and Technology

m_amine79@cmps2.iust.ac.ir

Abstract

CF-MARLOS is a cooperative federated intrusion-detection framework for B-NAT blockchain traffic that couples local CVAE-DQN training, Double-DQN policy learning, generator-based reconstruction, EVT-based open-set rejection, and learnable client aggregation. The system is built for horizontal federated learning under non-IID client partitions, so raw data remain local while the server coordinates shared representation learning. FedAvg and FedProx are used as standard baselines, while FMRL-LA provides a utility-aware two-phase aggregation strategy that prioritizes informative clients. After federated training, class-conditional EVT tails are fitted on validation reconstruction errors to calibrate a global unknown-rejection threshold. The proposed formulation targets three goals at once: preserve known-class performance, improve robustness under heterogeneous clients, and reject unseen attacks without destabilizing the closed-set classifier.

Keywords: federated learning intrusion detection open-set recognition EVT Double DQN blockchain traffic

1 Introduction

Blockchain-based traffic and other distributed networked environments produce data that are naturally siloed across nodes, organizations, or monitoring points. This makes centralized training inconvenient and often unrealistic. Federated learning provides a natural alternative, but standard aggregation rules such as FedAvg can degrade when local client distributions are highly non-IID. The problem becomes more difficult in intrusion detection, where the model must not only classify known attack categories but also reject previously unseen attacks.

The repository underlying this paper implements a cooperative federated multi-agent reinforcement learning pipeline that treats known-class intrusion detection as a CVAE-DQN problem and adds an EVT-based open-set rejection stage after training. The proposed system is built around four trainable blocks: a prior network, a

recognition network, a main Q-network, and a generator. A local target Q-network is retained for Double-DQN stability but is not federated as a separate object. The server aggregates shared parameters using FedAvg, FedProx, or FMRL-LA. Closed-set runs use the full label set, while open-set evaluation keeps FoT as the held-out unknown attack.

The main contributions are:

- a federated CVAE-DQN architecture for blockchain intrusion detection under non-IID client partitions;
- a generator-based reconstruction path that supports EVT open-set calibration;
- a utility-aware two-phase federated aggregation method, FMRL-LA, for heterogeneous clients;
- a complete evaluation protocol covering closed-set accuracy, open-set rejection, federated stability, and efficiency.

2 Related Work

Federated learning was introduced as a communication-efficient alternative to centralized learning in distributed settings [1]. FedProx extends this idea by adding a proximal penalty that limits client drift in heterogeneous networks [2]. In parallel, Double DQN reduces overestimation in Q-learning by decoupling action selection from target evaluation [3]. For open-set and unknown-class recognition, prior work has shown that confidence-based rejection is often insufficient, motivating tail modeling and EVT-style calibration [4, 5].

The present work combines these ideas in a single intrusion-detection pipeline. Rather than treating open-set rejection as an afterthought, reconstruction error is explicitly tied to the learned latent representation. Rather than assuming equal client value, the federated layer estimates client utility before aggregation. This is the key difference between CF-MARLOS and a standard federated classifier.

3 Problem Formulation

Let $\mathcal{D}_i = \{(x_j, y_j)\}$ denote the local dataset on client i , where x_j is a traffic feature vector and y_j is one of the known labels. The client set is $\{1, \dots, N\}$. The known-label set is

$$\mathcal{Y}_k = \{\text{Normal, BP, DoS, MitM}\},$$

while unseen attacks are mapped to an unknown label during evaluation.

The federated objective is to minimize the weighted sum of client losses:

$$\min_{\theta} \sum_{i=1}^N \frac{n_i}{\sum_j n_j} F_i(\theta),$$

where n_i is the number of local samples on client i . Under FedProx, the local objective becomes

$$F_i^{\text{prox}}(\theta) = F_i(\theta) + \frac{\mu}{2} \|\theta - \theta^{k-1}\|_2^2.$$

In FMRL-LA, the server instead applies utility-weighted aggregation:

$$\theta^k = \theta^{k-1} + \eta \frac{\sum_{i \in A_k} u_i (\theta_i^k - \theta^{k-1})}{\sum_{i \in A_k} u_i},$$

where A_k is the selected client set and u_i is the estimated utility of client i .

4 Proposed Methodology

We consider a horizontal federated setting in which each client trains locally on a private shard of B-NAT traffic. The server never receives raw samples. Instead, it receives model parameters and audit summaries, then performs aggregation and calibration in a controlled sequence. The full pipeline is illustrated in Figure 1.

4.1 System Overview

The system begins with raw B-NAT CSV traffic, which is preprocessed into tensors and partitioned into non-IID client shards. Clients perform local CVAE-DQN training, the server aggregates the resulting updates, and a global model is calibrated on validation reconstruction errors through EVT. This yields a thresholded open-set classifier that can preserve known-class performance while rejecting unknown attacks.

4.2 Local CVAE-DQN Agent

Each client contains a CVAE-DQN agent with four learned components: a prior network $p_{\theta}(z|s)$, a recognition network $q_{\phi}(z|s, a)$, a main Q-network $Q_{\psi}(z, s)$, and a generator $G_{\omega}(z, a)$. A local target Q-network Q_{ψ}^{-} is maintained only for temporal-difference stabilization. Replay memory and an epsilon-greedy policy support local interaction with the intrusion environment.

Algorithm 1 Federated CVAE-DQN Training and Open-Set Calibration

Require: Client datasets $\{D_i\}_{i=1}^N$, validation set D_{val} , rounds K , local configuration H , strategy $S \in \{\text{FedAvg, FedProx, FMRL-LA}\}$

Ensure: Final model θ^* , EVT models M_{evt} , rejection threshold δ

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1: Initialize  $\theta^0 = \{\theta_p^0, \theta_r^0, \theta_Q^0, \theta_G^0\}$ 
2: for  $k = 1$  to  $K$  do
3:   Server samples clients  $C_k$  and broadcasts  $\theta^{k-1}$ 
4:   for all  $i \in C_k$  in parallel do
5:      $\mu_i \leftarrow \mu$  if  $S = \text{FedProx}$  else 0
6:      $(\theta_i^k, m_i^k) \leftarrow \text{CLIENTLOCALUPDATE}(D_i, \theta^{k-1}, H, \mu_i)$ 
7:   end for
8:   if  $S = \text{FMRL-LA}$  then
9:      $\theta^k \leftarrow \text{FMRLLAUPDATE}(\theta^{k-1}, \{(\theta_i^k, m_i^k)\})$ 
10:  else
11:     $\theta^k \leftarrow \frac{\sum_{i \in C_k} n_i \theta_i^k}{\sum_{i \in C_k} n_i}$ 
12:  end if
13: end for
14:  $\theta^* \leftarrow \theta^K$ 
15:  $(M_{\text{evt}}, \delta) \leftarrow \text{EVT-CALIBRATION}(D_{\text{val}}, \theta^*)$ 
16: return  $\theta^*, M_{\text{evt}}, \delta$ 

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The local objective has three parts. The prior network is trained with a latent KL objective; the recognition and main Q-network are trained with a Double-DQN target; and, when FedProx is active, a proximal term constrains local drift from the broadcast global model. In compact form:

$$L_{\text{prior}} = \text{KL}(q_{\phi}(z|s, a^*) \parallel p_{\theta}(z|s)),$$

$$y = r + \gamma(1 - d) Q_{\psi}^{-}(z', s', \arg \max_a Q_{\psi}(z', s', a)),$$

$$L_{\text{TD}} = \text{Huber}(Q_{\psi}(z, s, a) - y),$$

$$L_{\text{prox}} = \frac{\mu}{2} \|\theta - \theta^{k-1}\|_2^2.$$

4.3 Generator-Based Reconstruction

The generator reconstructs an input feature vector from a latent code and a class condition:

$$\hat{x} = G_{\omega}(z, a).$$

It is trained only on correctly classified known samples so that the reconstruction error reflects the geometry of the known-class manifold rather than noise introduced by misclassified or unknown data. This error later becomes the basis for the open-set detector.

4.4 Federated Parameter Sharing

Only trainable parameters that define the shared representation are federated.

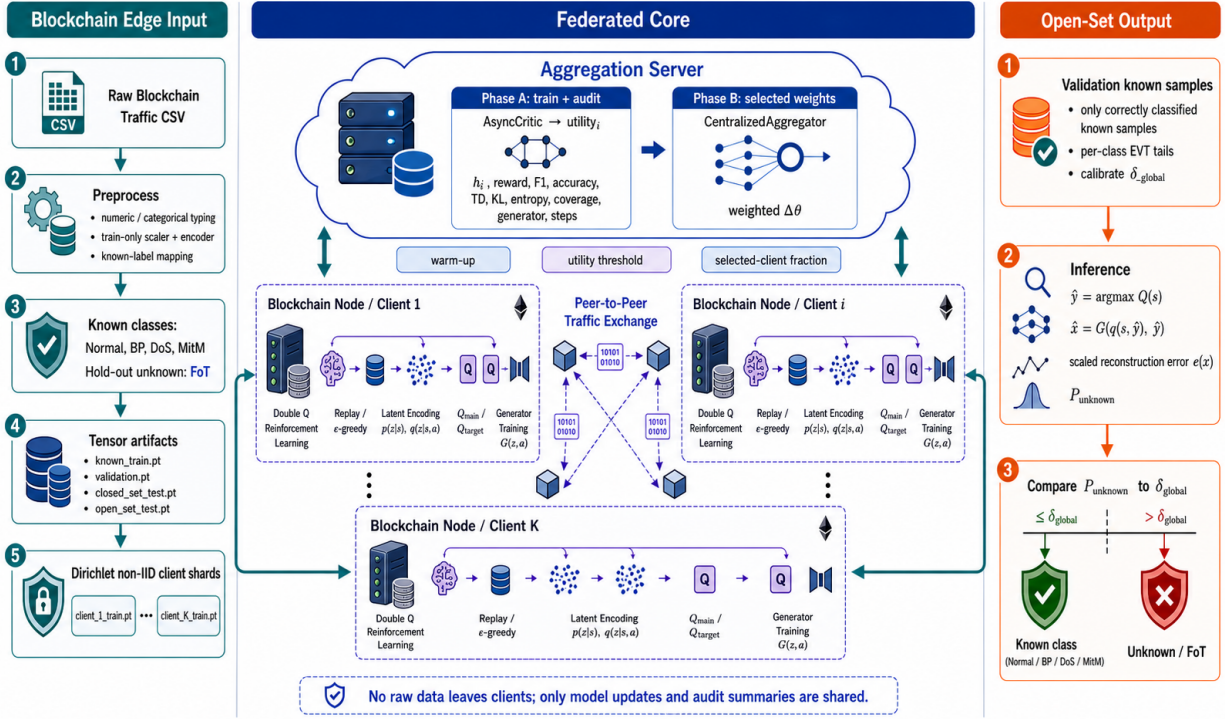


Figure 1: Overview of CF-MARLOS: preprocessing, non-IID client shards, local CVAE-DQN training, federated aggregation, and EVT-based open-set inference.

4.5 FedAvg and FedProx Baselines

FedAvg is the canonical baseline and uses sample-weighted averaging. FedProx keeps the same server aggregation rule but modifies the local objective with a proximal penalty. In this paper, FedAvg and FedProx are treated as baseline optimization strategies rather than separate algorithmic contributions.

4.6 FMRL-LA Cooperative Aggregation

FMRL-LA is the proposed cooperative aggregation mechanism. It uses two phases. In Phase A, clients train locally and upload audit metadata. The server estimates utility from latent summaries and diagnostics such as reward, accuracy, F1, TD stability, novelty, class entropy, label coverage, generator quality, and step count. In Phase B, selected clients upload cached weights and the server applies a utility-weighted update.

4.7 EVT Open-Set Calibration

After federated training, EVT is fitted on validation reconstruction errors that come only from correctly classified known samples. For each class, the upper tail of the reconstruction-error distribution is modeled with a generalized Pareto distribution. The resulting probability of unknownness is compared with a global threshold δ .

4.8 Algorithmic Summary

Algorithm 1 summarizes the end-to-end federated training and calibration flow. Algorithm 2 specifies the client-side CVAE-DQN update with the optional FedProx penalty. Algorithm 3 describes the utility-aware FMRL-LA aggregation rule. Algorithm 4 closes the loop by fitting EVT on validation reconstruction errors and applying the open-set decision rule at inference time.

5 Experimental Setup

The repository is scoped to B-NAT blockchain traffic, with the labels Normal, BP, DoS, and MitM as known classes and FoT as the held-out unknown attack. The final paper plan uses 10 clients and 100 logical federated rounds, although smaller development settings are retained in the repository for quick iteration.

The main baselines are centralized training, local-only training, FedAvg, FedProx, and FMRL-LA. For non-IID experiments, client shards are generated with a Dirichlet class partition. The main training hyperparameters are controlled through Hydra and include local episodes per round, replay-buffer size, batch size, Double-DQN settings, and generator training toggles.

Algorithm 2 Client-Side CVAE-DQN Local Update with Optional FedProx

Require: Local data D_i , broadcast parameters θ^{k-1} , local configuration H , proximal coefficient μ_i

Ensure: Updated parameters θ_i^k , diagnostics m_i^k

- 1: Load θ^{k-1} into p_θ , q_ϕ , Q_ψ , and G_ω
 - 2: Set $Q_\psi^- \leftarrow Q_\psi$ and store θ^{k-1} as the proximal reference
 - 3: Initialize replay buffer B_i and epsilon-greedy policy π_i
 - 4: **for** each local episode **do**
 - 5: Interact with the local environment and store transitions $(s_t, a_t, r_t, s_{t+1}, d_t, a_t^*)$ in B_i
 - 6: **if** $|B_i|$ is sufficient **then**
 - 7: Sample a mini-batch from B_i
 - 8: Update p_θ using L_{prior} and, if active, the FedProx penalty
 - 9: Update q_ϕ and Q_ψ using L_{TD} and, if active, the FedProx penalty
 - 10: Periodically soft-update or hard-copy Q_ψ^- from Q_ψ
 - 11: **end if**
 - 12: **end for**
 - 13: Build the set of correctly classified known samples
 - 14: **if** generator training is enabled **then**
 - 15: Train G_ω only on correctly classified known samples
 - 16: **end if**
 - 17: **return** $\theta_i^k = \{\theta_p, \theta_r, \theta_Q, \theta_G\}$ and diagnostics m_i^k
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6 Evaluation Metrics

Closed-set evaluation uses accuracy, macro F1, balanced accuracy, and per-class confusion matrices. Open-set evaluation uses AUROC, AUPRC, FPR@95%TPR, unknown F1, unknown detection rate, and known accuracy after rejection. Federated evaluation uses convergence speed, selected-client fraction, stability across rounds, communication cost, and round-level utility. Efficiency is reported through runtime and cumulative transmitted model bytes.

7 Planned Results and Discussion

The final manuscript should report three comparisons: closed-set performance, open-set rejection, and federated robustness under non-IID data. In particular, the results section should compare FedAvg, FedProx, and FMRL-LA under matched client partitions and identical training budgets. The paper should also include communication-cost curves and round-wise stability plots so that the gain from utility-aware aggregation can be interpreted against its overhead.

Table 1: Federated and non-federated components

Federated	Role
Prior network $p_\theta(z s)$	Shared latent structure
Recognition network $q_\phi(z s, a)$	Latent inference and reconstruction
Main Q-network $Q_\psi(z, s)$	Global known-class policy
Generator $G_\omega(z, a)$	Shared reconstruction behavior
Not federated	Role
Target Q-network Q_ψ^-	Local stabilization copy
Replay buffer	Local trajectory history
Optimizer state	Private training state
EVT models and thresholds	Post-training calibration artifacts
Raw client data	Remains local for privacy

8 Ablation Study

The core ablations are: removing EVT, removing the generator, replacing FMRL-LA with FedAvg, replacing FMRL-LA with FedProx, and sweeping the EVT tail size and utility threshold. These ablations isolate the contribution of open-set calibration, reconstruction learning, and client selection.

9 Limitations and Threats to Validity

The main limitations are calibration sensitivity, the cost of two-phase federated coordination, and the current focus on B-NAT rather than multiple external datasets. EVT quality depends on validation splits, and FMRL-LA requires additional server-side state. These limitations should be stated explicitly in the final paper.

10 Reproducibility

The repository provides Hydra-driven configuration, seed control, checkpointing, local artifact export, and run-level plots. These features make the pipeline reproducible across preprocessing, training, evaluation, and figure generation. The manuscript should refer to the stored outputs for exact benchmark replication.

11 Conclusion

CF-MARLOS combines local CVAE-DQN learning, federated aggregation, generator-based reconstruction, and EVT open-set calibration into a single intrusion-detection framework. FedAvg and FedProx provide standard baselines, while FMRL-LA adds utility-aware client selection for heterogeneous data. The resulting pipeline is suitable for known-class classification, unknown attack rejection, and federated learning under non-IID client data.

Algorithm 3 FMRL-LA Utility-Aware Client Selection and Aggregation

Require: Previous global parameters θ^{k-1} , client updates and diagnostics $\{(\theta_i^k, m_i^k)\}$, utility threshold λ , minimum selected clients $q_{\min}$, maximum fraction $\rho_{\max}$, step size η

Ensure: Updated global parameters θ^k

- 1: Estimate utility u_i for each client using latent summaries and diagnostics
 - 2: Select $A_k = \{i : u_i \geq \lambda\}$
 - 3: **if** $|A_k| < q_{\min}$ **then**
 - 4: Add the highest-utility clients until $|A_k| = q_{\min}$
 - 5: **end if**
 - 6: Limit $|A_k| \leq \lceil \rho_{\max} |C_k| \rceil$
 - 7: **for all** $i \in A_k$ **do**
 - 8: $\Delta_i \leftarrow \theta_i^k - \theta^{k-1}$
 - 9: **end for**
 - 10: $\theta^k \leftarrow \theta^{k-1} + \eta \frac{\sum_{i \in A_k} u_i \Delta_i}{\sum_{i \in A_k} u_i}$
 - 11: Update server-side critics and mixer using round-level utility feedback
 - 12: **return** θ^k
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References

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Algorithm 4 EVT Calibration and Open-Set Inference

Require: Validation set D_{val} , trained model θ^* , tail fraction α_{evt} , target known false-positive rate β

Ensure: EVT models M_{evt} , global threshold δ , final prediction $\hat{y}$

- 1: Collect reconstruction errors for correctly classified known validation samples
 - 2: **for all** known classes c **do**
 - 3: Fit an EVT tail model to the upper tail of the errors for class c
 - 4: **end for**
 - 5: Estimate validation unknown probabilities and set δ from the $(1 - \beta)$ quantile
 - 6: **for** each inference sample x **do**
 - 7: Predict $\hat{c} = \arg \max_c Q_\psi(p_\theta(x), x, c)$
 - 8: Compute reconstruction error e_x using q_ϕ and G_ω
 - 9: Compute $p_u = M_{\text{evt}}[\hat{c}](e_x)$
 - 10: **if** $p_u \geq \delta$ **then**
 - 11: $\hat{y} \leftarrow \text{Unknown}$
 - 12: **else**
 - 13: $\hat{y} \leftarrow \hat{c}$
 - 14: **end if**
 - 15: **end for**
 - 16: **return** $M_{\text{evt}}, \delta, \hat{y}$
-